

Lesson 29 — Bias, Fairness & Responsible Use

Matches LEGO Teacher Portal lesson: "Super Speed Skateboard".

Objective (I can...): I can identify how biased data causes unfair AI, test for it in my own model, and argue for responsible use. **CS concept:** Dataset bias and its impacts; privacy and responsible use · **Builds on:** Lesson 28 (data quality). *Privacy: on-device in the browser, no login, images not saved — and this lesson makes that a design value, not just a fact.*

AI Connection (2 min): Real AI systems have made unfair decisions because their training data didn't represent everyone — this is a top concern for working engineers. *First-principles question:* "If your classifier only ever saw examples that look like you, who might it fail for, and whose responsibility is that?"

Materials: Device per pair (Teachable Machine), LEGO CS & AI kit, notebooks, a short "bias scenario" card set. **Model reuse:** AI responder model from Lesson 26 — test and retrain.

60-minute flow - Warm-up (8): "Who's missing?" — pairs read a short scenario (a classifier trained on only one kind of example) and predict who it will fail for and the real-world harm. - **Concept + demo (10):** Demo (or set up) a deliberately biased model — e.g., trained in only bright light or on only one object orientation — and show it fail on the under-represented case. Name *bias* as a data problem with people-impacts, and connect to the privacy choices (on-device, nothing saved) as responsible design. - **Build/Code (30):** Pairs deliberately build a biased version of their classifier (narrow examples), document its failure on a held-out case, then fix it by broadening the dataset and re-testing. They write a short notebook entry: the bias, who it would affect, and the fix. *Swap at 15 min.* - **Challenge (7):** Draft two "responsible-use rules" for a product built on their model (e.g., confidence gate, human override, what data it should and shouldn't keep). - **Share + discussion (5):** Pairs present their biased-vs-fixed result and read one responsible-use rule. Discuss the privacy tradeoff: convenience vs. keeping data private. Model stays built.

Pair roles: Driver builds the biased/fixed datasets, Navigator documents the impact and the rules. Swap at 15 min. (*Trio: third runs the held-out test.*)

Differentiation: *Support* — provide the bias scenario and one rule stem to complete. *Extension / fast-finisher* — analyze a second, subtler bias and propose a test for it. *Reteach* — contrast two trainings side by side (narrow vs. broad) so the effect is concrete.

Exit check: Pair demonstrates a biased model, fixes it with better data, and can state who biased AI harms and one responsible-use rule. **At-home (optional):** Find or recall a news example of unfair or privacy-invasive AI; write what data choice likely caused it. ~10 min.